Test Series No.

1.6

MASTERY TEST SERIES

DURATION: 90 Minutes | **M. MARKS: 400**

DATE: 24/02/2026

SYLLABUS

Physics:	Wave Optics, Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor Electronics - Materials, Devices and Simple Circuits + 30% Questions from cumulative syllabus of previously conducted part test
Chemistry:	Coordination Compounds, The p-Block Elements, The d- and f- block Elements, Salt analysis + 30% Questions from cumulative syllabus of previously conducted part test
Botany:	Organisms and Population, Ecosystem, Biodiversity and Conservation + 30% Questions from cumulative syllabus of previously conducted part test
Zoology:	Biotechnology - Principles and Processes, Biotechnology and its Applications + 30% Questions from cumulative syllabus of previously conducted part test

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **90 minutes** duration and the Test Booklet contains **100** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 400.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: For nuclear fusion reactions, very high temperature is required.

Reason R: Only at high temperature, the fusing nuclei get enough energy to overcome electrostatic force of repulsion between them.

In the light of the above statements, choose the **correct** answer from the options given below:

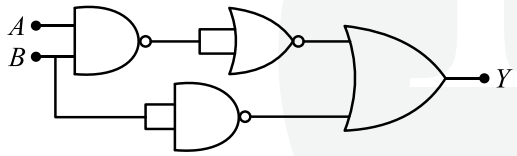
(1) A is true but R is false

(2) A is false but R is true

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

2. The output Y of the logic circuit shown in figure will be:



(1) $\left(\overline{A \cdot B}\right) + \overline{B}$

(2) $(A \cdot B) + \overline{B}$

(3) $A \cdot B$

(4) B

3. The total electric flux through a closed surface enclosing an α-particle is: (e is charge on a proton)

(1) Zero (2) $\frac{2e}{\varepsilon_0}$

(3) $\frac{4e}{\varepsilon_0}$ (4) $\frac{3e}{\varepsilon_0}$

4. If a half-wave rectifier is used to convert n Hz input AC voltage into DC, then the frequency (in Hz) of the rectified output voltage is:

(1) n (2) $\frac{n}{2}$

(3) $2n$ (4) $4n$

5. In Young's double slit experiment, the phase difference (in radian) between the light waves reaching 5th dark fringe will be (wavelength of light $\lambda = 4500$ Å)

(1) 5π (2) 7π

(3) 9π (4) 3π

6. When YDSE is conducted with white light, a white fringe is observed at the centre of the screen. When the screen is moved away perpendicularly from the slits by another 5 mm, then at centre of screen, fringe

(1) remain white

(2) becomes red

(3) becomes yellow

(4) becomes violet

7. A circular loop of single turn of radius r is placed in a region where magnetic field perpendicular to plane of loop increases with respect to time as $B(t) = (at)$tesla then magnitude of induced emf (in volts) in coil is:

(1) $\pi r^2 a$

(2) $3\pi r^2 a$

(3) $2\pi r^2 a$

(4) $4\pi r^2 a$

8. In the 3rd orbit of He^+ (Helium) ion, the speed of electron will be:

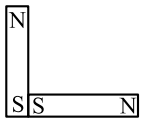
(1) 1.46×10^6 m/s

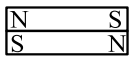
(2) 0.73×10^6 m/s

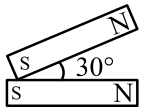
(3) 3.0×10^8 m/s

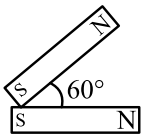
(4) 2.92×10^6 m/s

9. Following figures show the arrangement of bar magnets with different configurations. Each bar magnet has magnetic dipole moment $\vec{m}$. Which configuration has highest net magnetic dipole moment?

A. 

B. 

C. 

D. 

(1) A (2) B

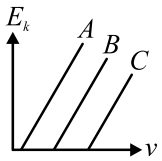
(3) C (4) D

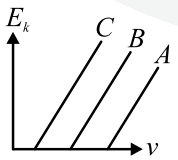
10. A polarizer and an analyser are kept such that maximum amount of light is transmitted through them. Intensity of that transmitted light is I_0. When the analyser is rotated through an angle of 60°, then the intensity of light transmitted is:

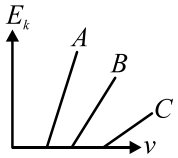
(1) $\frac{2}{3}I_0$ (2) $\frac{I_0}{2}$

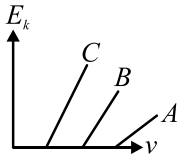
(3) $\frac{I_0}{4}$ (4) $\frac{3}{4}I_0$

11. The work functions of three metals A, B and C are W_A, W_B and W_C respectively such that $W_A > W_B > W_C$. The most appropriate graph between maximum kinetic energy of emitted photoelectron and frequency v of incident radiation is:

(1) 

(2) 

(3) 

(4) 

12. The de-Broglie wavelength of a neutron when its kinetic energy E_k is λ. Its de-Broglie wavelength when its kinetic energy is $4E_k$ is:

(1) 4λ (2) 2λ

(3) $\frac{\lambda}{2}$ (4) $\frac{\lambda}{4}$

13. The binding energy per nucleon of a hydrogen nucleus is:

(1) 13.6 eV

(2) zero

(3) infinity

(4) 8.5 MeV

14. The speed of electromagnetic wave in a medium which has relative magnetic permeability 2 and relative electric permittivity 8 is:

(1) 1.5×10^8 m/s

(2) 1×10^8 m/s

(3) 2.5×10^8 m/s

(4) 7.5×10^7 m/s

15. Choose the **correct** statement?

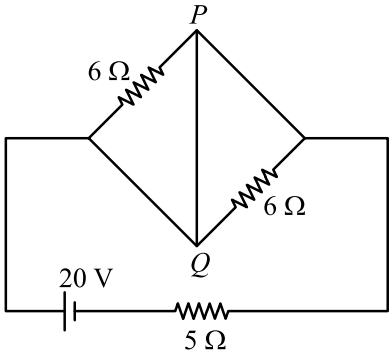
(1) Resonance condition is possible in series RL and series RC circuit.

(2) Transformer is a device which primarily uses principle of self induction.

(3) In a purely inductive or capacitive circuit, the current flowing in the circuit is referred as wattless current.

(4) In a step-up transformer, primary coil has a greater number of turns than the secondary coil.

16. The value of current flowing through wire PQ as shown in the figure is:



(1) Zero

(2) 3 A

(3) 4 A

(4) 1.2 A

[Contd...

17. If photons of energy 1 eV and 2.5 eV respectively are incident on a given metallic plate of work function 0.5 eV, one after another, then ratio of maximum kinetic energy of photoelectrons emitted by them respectively will be:

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$

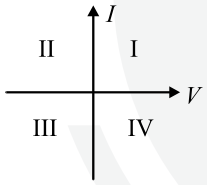
(3) $\frac{1}{4}$ (4) $\frac{1}{6}$

18. The binding energy per nucleon for deuterium and helium nuclei are 1.1 MeV and 7.0 MeV respectively. When two deuterium nuclei fuse to form a helium nucleus, then the energy released in the fusion process is:

(1) 2.2 MeV (2) 28.0 MeV

(3) 30.2 MeV (4) 23.6 MeV

19. Choose the **correct** statement regarding *I-V* characteristics of special purpose P-N junction diodes?



A. For photodiode, the *I-V* characteristic lies in III quadrant of given graph.

B. For Zener-diode (reverse-bias state), the *I-V* characteristic lies in III quadrant of given graph.

C. For solar cell, the *I-V* characteristics lies in IV quadrant of given graph.

(1) A and C only (2) B and C only

(3) A and B only (4) A, B and C

20. The dimensional formula of frequency is same as the dimensional formula of:

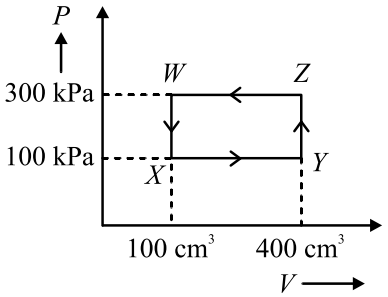
(1) angular momentum

(2) velocity

(3) ratio of planck's constant and moment of inertia.

(4) product of mass and elastic modulus.

21. A thermodynamic system is taken through cycle *WXYZ* as shown. The work done by the gas along the path *WXY* is:



(1) 0 (2) 30 Joule

(3) 20 Joule (4) 10 Joule

22. For a hydrogen like species with atomic number *Z*, ratio of magnitude of angular momentum for electron revolving in 2nd excited state and 4th excited state is:

(1) 3 : 4 (2) 2 : 3

(3) 1 : 2 (4) 3 : 5

23. Light of wavelength 6328 Å is incident normally on a single slit having a width of 0.2 mm. The angular width of the central maximum measured from minima of one side to minima of other side of diffraction pattern formed on a screen which is 9 m away will be nearly;

(1) 0.36 degree (2) 0.18 degree

(3) 0.72 degree (4) 0.09 degree

24. Ratio of longest wavelength of Lyman series to shortest wavelength of Lyman series in hydrogen spectrum is;

(1) $\frac{1}{3}$ (2) $\frac{3}{4}$

(3) $\frac{4}{3}$ (4) $\frac{1}{4}$

25. A magnetic force acting on a charged particle is given by $(-8\text{N}\hat{k})$. The velocity of the particle is $(2\hat{i}+3\hat{j})\times10^6$ m/s and the magnetic field of 2 T is applied in the +*y* direction. Find the charge on the particle.

(1) $2\mu C$ (2) $3\mu C$

(3) $-2\mu C$ (4) $-3\mu C$

[Contd...

26. Match List-I with List-II.

List-I (Transition metal ion)		List-II (Spin only magnetic moment (B.M.))	
(A)	Ti^{3+}	(I)	3.87
(B)	V^{2+}	(II)	0.00
(C)	Ni^{2+}	(III)	1.73
(D)	Sc^{3+}	(IV)	2.84

Choose the **correct** answer from the options given below:

(1) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(2) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

(3) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(4) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

27. Given below are two statements:

Statement I: A catalyst catalyses both the spontaneous and non-spontaneous reactions.

Statement II: The term $e^{-\frac{E_a}{RT}}$ represents the fraction of total number of molecules having energies equal or greater than E_a.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

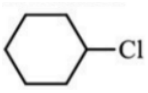
(1) Both Statement I and Statement II are correct.

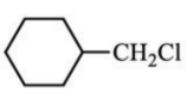
(2) Statement I is correct but Statement II is incorrect.

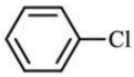
(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

28. Which of the following halides undergoes S_N2 reaction fastest?

(1) 

(2) 

(3) 

(4) $CH_2 = CH - Cl$

29. The only well characterised compound of Bi(V) is:

(1) BiI_5 (2) $BiCl_5$

(3) BiF_5 (4) $BiBr_4$

30. What will be the **correct** order for the wavelengths of absorption in the visible region for the following:

I. $[CoCl(NH_3)_5]Cl_2$

II. $[Co(NH_3)_3(H_2O)_3]Cl_3$

III. $[Co(NH_3)_6]Cl_3$

(1) III > II > I (2) I > II > III

(3) II > I > III (4) II > III > I

31. **Incorrect** statement among the following is:

(1) liquid helium is used as cryogenic liquid

(2) neon is used in discharge tubes and fluorescent bulbs for advertisement display purposes.

(3) argon is used for electric bulbs.

(4) xenon is used in filling balloons for meteorological observations

32. The **correct** order of basicity of oxides of vanadium is

(1) $V_2O_3 > V_2O_4 > V_2O_5$

(2) $V_2O_3 > V_2O_5 > V_2O_4$

(3) $V_2O_5 > V_2O_4 > V_2O_3$

(4) $V_2O_4 > V_2O_3 > V_2O_5$

33. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: $[(Co)_2(CO)_8]$ is an organometallic compound.

Reason R: It involves synergic bonding.

In the light of the above statements, choose the **correct** answer from the options given below:

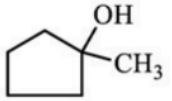
(1) Both A and R are true and R is the correct explanation of A.

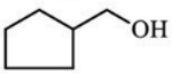
(2) Both A and R are true but R is NOT the correct explanation of A.

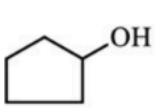
(3) A is true but R is false.

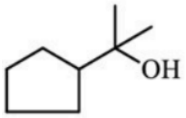
(4) A is false but R is true.

34. Which of the following alcohols give ketone as major product when their vapours are passed over heated copper at 573 K?

(1) 

(2) 

(3) 

(4) 

35. Identify the **incorrect** statements about group 15 elements:

(A) Dinitrogen is a diatomic gas which acts like an inert gas at room temperature.

(B) The common oxidation states of these elements are -3, +3 and +5.

(C) Nitrogen has unique ability to form pπ-pπ multiple bonds.

(D) The stability of +5 oxidation states increases down the group.

(E) Nitrogen shows a maximum covalency of 6.

Choose the **correct** answer from the options given below.

(1) A, B, D only (2) A, C, E only

(3) B, D, E only (4) D and E only

36. Molybdenum (M_O) and rhodium (Rh) have "x" and "y" number of electrons in their respective 4d orbitals. The value of $x + y$ is _____. (Atomic no. $\Rightarrow M_O = 42$ and Rh = 45)

(1) 13 (2) 12

(3) 10 (4) 8

37. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: Among group-16 hydrides acidic character increases from H_2O to H_2Te.

Reason R: As the size of M increases, bond energy of M-H bond decreases. (M = elements of 16 group)

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

38. Given below are two statements:

Statement I: $[Co(NH_3)_4Cl_2]^+$ is a homoleptic complex ion.

Statement II: Oxidation number of copper in $[Cu(CN)_4]^{3-}$ is +1.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I and Statement II both are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Statement I and Statement II both are incorrect.

39. Match List-I with List-II.

List-I		List-II	
(A)	Second order reaction	(I)	Natural radioactive decay
(B)	Zero order reaction	(II)	Unit of rate constant is $M^{-1} s^{-1}$
(C)	First order reaction	(III)	Inversion of cane sugar
(D)	Pseudo first order reaction	(IV)	Thermal decomposition of HI on gold surface

Choose the **correct** answer from the options given below:

(1) A-III; B-II; C-IV; D-I

(2) A-II; B-I; C-IV; D-III

(3) A-II; B-IV; C-I; D-III

(4) A-I; B-III; C-IV; D-II

40. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Aryl halides are highly reactive towards nucleophilic substitution reaction.

Reason R: In aryl halides, C-Cl bond acquires a partial double bond character.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false

(2) A is false but R is true

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

41. Which of the following statements are **correct** about Zn, Cd and Hg?

A. They exhibit high enthalpy of atomisation as the d-subshell is full.

B. Zn and Cd do not show variable oxidation state while Hg shows +I and +II.

C. Compounds of Zn, Cd and Hg are paramagnetic in nature.

D. Zn, Cd and Hg are called soft metals.

Choose the *most appropriate* from the options given below:

(1) B, D only (2) B, C only

(3) A, D only (4) C, D only

[Contd...

42. N-O-N bond is present in:

(1) N_2O_3

(2) N_2O_5

(3) N_2O_4

(4) N_2O

43. Which among the following can exist as facial and meridional isomers?

(1) $[Co(NH_3)_3(NO_2)_3]$

(2) $[CoCl_2(en)_2]^+$

(3) $[Co(en)_3]Cl_3$

(4) $[Co(NH_3)_4Cl_2]Cl$

44. Match the List-I with List-II:

List-I (Cations)		List-II (Group reaction)	
(A)	Pb^{2+}, Cu^{2+}	(I)	H_2S gas in presence of dilute HCl
(B)	Al^{3+}, Fe^{3+}	(II)	$(NH_4)_2CO_3$ in presence of NH_4OH
(C)	Co^{2+}, Ni^{2+}	(III)	NH_4OH in presence of NH_4CI
(D)	Ba^{2+}, Ca^{2+}	(IV)	H_2S in presence of NH_4OH

(1) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

(2) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(3) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

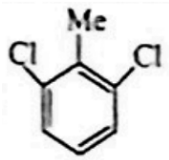
(4) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

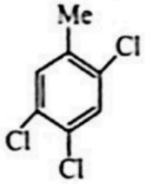
45. Highest melting point is of:

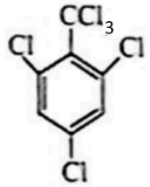
(1) H_2Te (2) H_2S

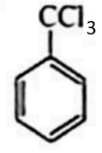
(3) H_2O (4) H_2Se

46. By passing excess of $Cl_2(g)$ in boiling toluene, which one of the following compounds is exclusively formed?

(1) 

(2) 

(3) 

(4) 

47. The orange colour of $K_2Cr_2O_7$ and purple colour of $KMnO_4$ is due to

(1) Charge transfer transition in both.

(2) d→d transition in $KMnO_4$ and charge transfer transitions in $K_2Cr_2O_7$

(3) d→d transition in $K_2Cr_2O_7$ and charge transfer transitions in $KMnO_4$.

(4) d→d transition in both

48. Hybridization of $[NiCl_4]^{2-}$ is:

(1) sp^3d

(2) sp^2

(3) dsp^2

(4) sp^3

49. When a solution of mixture having two inorganic salts was treated with freshly prepared ferrous sulphate in acidic medium, a dark brown ring was formed whereas on treatment with neutral $FeCl_3$, it gave deep red colour which disappeared on boiling and a brown red ppt was formed. The mixture contains

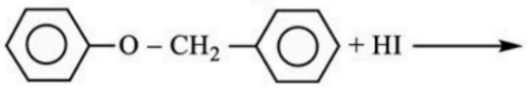
(1) $CH_3COO^- \ \& \ NO_3^-$

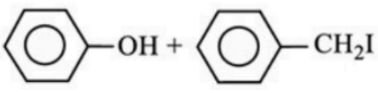
(2) $C_2O_4^{2-} \ \& \ NO_3^-$

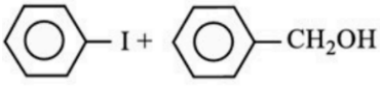
(3) $SO_3^{2-} \ \& \ CH_3COO^-$

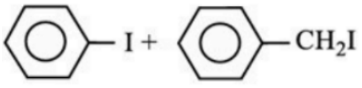
(4) $SO_3^{2-} \ \& \ C_2O_4^{2-}$

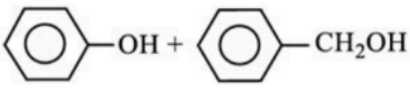
50. The products of the given reaction are:



(1) 

(2) 

(3) 

(4) 

[Contd...

51. Which of the following statements is **correct** about centrosomes?
(1) They undergo duplication during the S phase in plant cells.
(2) They move to opposite poles during metaphase.
(3) They radiate microtubules during prophase.
(4) They are membrane-bound organelles.

52. How many of the following are natural terrestrial ecosystems?
Estuary, Wetland, Desert, Crop field, Forest, Aquarium
(1) One (2) Two
(3) Three (4) Four

53. Which of the following equations **correctly** represents the Verhulst-Pearl logistic growth model?
(1) $\frac{dN}{dt} = rN$ (2) $\frac{dN}{dt} = rN\frac{(K-N)}{K}$
(3) $N_t = N_0 e^{rt}$ (4) $\frac{dN}{dt} = rN\left(\frac{K}{N}\right)$

54. Which of the following statements is **incorrect** about Carolus Linnaeus?
(1) He proposed binomial nomenclature.
(2) He included bacteria, blue-green algae, fungi, mosses, ferns, gymnosperms, and angiosperms under the group 'Plants'.
(3) He grouped *Chlamydomonas* and *Spirogyra* under algae.
(4) He gave a natural system of classification based on the structure of the androecium.

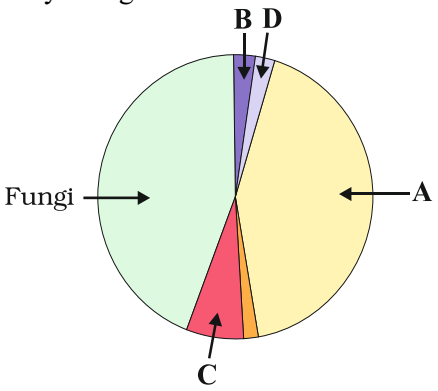
55. The medicinal plant *Rauwolfia vomitoria*, growing in different Himalayan ranges, shows variation in the concentration and potency of reserpine. This represents:
(1) Species diversity
(2) Genetic diversity
(3) Ecological diversity
(4) Community diversity

56. Match **List-I** with **List-II**

List-I (Key Enzyme/Molecule)		List-II (Associated process)	
(A)	RuBisCO	(I)	Glycolysis
(B)	PEPcase	(II)	Krebs cycle
(C)	Hexokinase	(III)	C_3 cycle
(D)	Citrate synthase	(IV)	C_4 cycle

Choose the *most appropriate* answer from the options given below:
(1) A–III, B–IV, C–I, D–II
(2) A–I, B–II, C–III, D–IV
(3) A–IV, B–III, C–II, D–I
(4) A–III, B–I, C–IV, D–II

57. Given below is a pie chart representing the proportionate number of species of major plant taxa. Identify the label that represents the species diversity of algae.



(1) B (2) C
(3) D (4) A

58. Which of the following parasites requires two intermediate hosts (a snail and a fish) to complete its life cycle?
(1) Malarial parasite
(2) Human Liver fluke
(3) Copepodes
(4) *Cuscuta*

59. Which of the following features is generally common between monocot and dicot plants?
(1) Type of cells present in the hypodermis of the stem
(2) Type of venation
(3) Presence of endosperm in the mature seed
(4) Orientation of protoxylem and metaxylem in the root

60. If energy available at producer level is 4000 J, then how much energy is typically available to the primary carnivore?
(1) 400 J (2) 20 J
(3) 4 J (4) 40 J

61. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Insertion or deletion of three or its multiple bases changes the reading frame from the point of insertion or deletion.
Reason R: Deletions and insertions of base pairs of DNA, causes frame-shift mutations.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is NOT the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

[Contd...

62. Strict protection of all hotspots could reduce the ongoing mass extinctions by almost
(1) 30 per cent (2) 50 per cent
(3) 75 per cent (4) 90 per cent

63. Which of the following interactions is characterized by '–, 0'?
(1) Predation (2) Amensalism
(3) Commensalism (4) Competition

64. Cleistogamous flowers are advantageous because they:
(1) encourage cross-pollination.
(2) produce assured seed-set even without pollinators.
(3) lead to high genetic variation.
(4) open to expose anthers and stigma.

65. Humus is further degraded by some microbes and release of inorganic nutrients occur by the process called:
(1) catabolism. (2) fragmentation.
(3) leaching. (4) mineralisation.

66. A bioactive molecule that is used as blood cholesterol lowering agent is obtained from:
(1) *Trichoderma polysporum*
(2) *Monascus purpureus*
(3) *Propionibacterium sharmanii*
(4) *Saccharomyces cerevisiae*

67. Match **List-I** with **List-II:**

List-I (Country)		**List-II (Number of bird species)**	
(A)	Colombia	(I)	105
(B)	New York	(II)	More than 1,200
(C)	Greenland	(III)	Nearly 1,400
(D)	India	(IV)	56

Choose the *most appropriate* answer from the options given below:
(1) A-III, B-I, C-IV, D-II
(2) A-II, B-IV, C-I, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-I, B-II, C-III, D-IV

68. Given below are two statements.
Statement I: Predation is nature's way of transferring to higher trophic levels the energy fixed by plants.
Statement II: The intrinsic rate of natural increase is a measure of the inherent potential of a population to grow.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement 1 is incorrect but Statement II is correct.

69. Cytokinins were first discovered as 'kinetin' from:
(1) corn kernels.
(2) coconut milk.
(3) autoclaved herring sperm DNA.
(4) human urine.

70. In which of the following food chains is the ecological pyramid inverted?
(1) Grassland ecosystem – Pyramid of numbers
(2) Tree ecosystem – Pyramid of energy
(3) Sea ecosystem – Pyramid of energy
(4) Sea ecosystem – Pyramid of biomass

71. Which of the following is **incorrect** about biodiversity in India?
(1) India contributes 8.1% to global species diversity.
(2) India has nearly 1,00,000 species of plants that are already discovered.
(3) India has more than 50,000 genetically different strains of rice.
(4) More than 650 species currently face the threat of extinction.

72. Which of the following is **not** a defence mechanism evolved by plants against herbivores?
(1) Thorns in *Acacia*
(2) Production of cardiac glycosides in *Calotropis*
(3) Production of nicotine, caffeine, and quinine
(4) Showing sexual deceit

73. Which of the following taxonomic groups has the highest number of species among animals?
(1) Fish (2) Insects
(3) Reptiles (4) Mammals

74. Which of the following is **incorrect** about the detritus food chain?
(1) It forms the major conduit for energy flow in terrestrial ecosystems.
(2) The fraction of energy flow through it is greater than that through the grazing food chain in aquatic ecosystems.
(3) It begins with dead organic matter.
(4) Organisms of this food chain secrete digestive enzymes that break down detritus

75. Which of the following are invasive alien species in India?
A. Nile Perch
B. *Parthenium*
C. *Eichhornia*
D. *Lantana*
E. *Clarias gariepinus*
Choose the **correct** answer from the options given below:
(1) A, B and C only
(2) B, C, D and E only
(3) A, B, C, D and E
(4) A, B and D only

[Contd...

76. Identify the **incorrect** statement regarding restriction enzymes:
(1) They belong to a larger class of enzymes called nucleases.
(2) Exonucleases remove nucleotides from the ends of the DNA.
(3) Endonucleases make cuts at specific positions within the DNA.
(4) Hind II was the first restriction endonuclease characterized, and it cuts DNA by recognizing a specific sequence of five base pairs.

77. The "Bt" in Bt cotton stands for which of the following?
(1) Biotechnology-based cotton
(2) Barium-treated seeds
(3) *Bacillus thuringiensis*
(4) Biolistic transformation

78. At which gene Pvu II restriction site is present in pBR322 cloning vector?
(1) amp^R
(2) tet^R
(3) ori
(4) rop

79. Which of the following biotechnological products should go through clinical trial before human use?
(1) Bt cotton
(2) Golden rice
(3) Recombinant insulin
(4) Flavr Savr tomato

80. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In some children, ADA deficiency can be permanently cured if treated at early embryonic stages.
Reason R: Lymphocytes from the blood of the patient are immortal and do not require periodic infusion.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

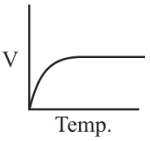
81. "Rosie," the first transgenic cow produced in 1997, was significant because her milk contained:
(1) Human insulin-like growth factor
(2) High amounts of cholesterol
(3) Recombinant alpha-1-antitrypsin
(4) Human alpha-lactalbumin

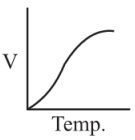
82. Match **List-I** with **List-II**.

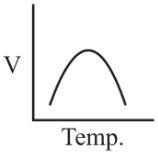
List-I		List-II	
(A)	Cotton bollworms	(I)	*cry*IAb
(B)	Corn borer	(II)	*cry*IAc and *cry*IIAb
(C)	Tobacco roots	(III)	*Meloidegyne incognitia*
(D)	Lepidopterans	(IV)	Tobacco budworm

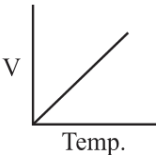
Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-IV, D-III
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-II, C-IV, D-I

83. Which of the following **correctly** represent a disposable barrier, a reusable barrier and a hormonal contraceptive method respectively?
(1) Diaphragm, Cervical cap and oral contraceptive pills
(2) Vaults, Nirodh and Implants
(3) Cervical cap, condom and Multiload-375
(4) Condom, Vaults and LNG-20

84. Which of the given graphs, shows the effect of temperature on the velocity of a typical enzyme catalysed reaction?
(1) 

(2) 

(3) 

(4) 

85. Regarding the naming of *Eco*RI, which of the following statement is **correct**?
(1) The letter 'R' is derived from the genus name.
(2) The first letter 'E' comes from the species name.
(3) Roman numbers indicate the order in which enzymes were isolated from that strain of bacteria.
(4) 'co' represents the name of the strain RY 13.

[Contd...

86. Motile zygote of *Plasmodium* occurs in:
(1) gut of female *Anopheles*
(2) salivary glands of *Anopheles*
(3) human RBCs
(4) human liver

87. Match **List-I** with **List-II**.

List-I (Technique)		List-II (Description)	
(A)	Electrophoresis	(I)	*In vitro* DNA amplification
(B)	Elution	(II)	Introducing DNA into a host
(C)	Transformation	(III)	Extracting DNA from agarose gel
(D)	PCR	(IV)	Separation of DNA fragments

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-II, C-III, D-I

88. In the context of molecular diagnosis, what is a probe?
(1) A radioactive double-stranded DNA molecule
(2) A single-stranded DNA or RNA tagged with a radioactive molecule
(3) An antibody used in ELISA
(4) A vector used for gene therapy

89. Which of the following assisted reproductive technology helps in both male infertility (due to low sperm count) and female infertility (due to blocked fallopian tubes)?
(1) Artificial insemination
(2) Gamete intrafallopian transfer
(3) Intra uterine insemination
(4) Intra cytoplasmic sperm injection

90. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: DNA cannot pass through cell membranes easily.
Reason R: DNA is a hydrophobic molecule therefore it cannot cross the plasma membrane easily.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

91. Which of the following technique is used to detect mutations in genes in suspected cancer patients?
(1) ELISA (2) Serum analysis
(3) PCR (4) Urine analysis

92. Consider the following statements regarding PCR and choose the option with **correct** ones only.
A. It stands for Polymerase Chain Reaction.
B. Multiple copies of DNA are synthesized *in vivo*.
C. It uses two sets of primers and DNA ligase.
D. *Taq* polymerase is isolated from the bacterium *Thermus aquaticus*.
E. The process involves three steps: Denaturation, Annealing, and Extension.
(1) A, C, D and E only
(2) B, C and D only
(3) A, D and E only
(4) A, B, C, D and E

93. How many documented varieties of Basmati rice are grown in India?
(1) 200,000 (2) 30
(3) 12 (4) 27

94. During the visualization of DNA fragments in gel electrophoresis, a student observes bright orange-coloured bands. Which of the following specific conditions led to this observation?
(1) The gel was stained with methylene blue and viewed under visible light.
(2) The gel was stained with ethidium bromide and viewed under visible light.
(3) The gel was stained with ethidium bromide and exposed to UV radiation.
(4) The DNA absorbs UV radiation naturally and glows orange without staining.

95. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Carcinogens	(I)	Helps in destroying the tumor
(B)	MRI	(II)	Tumor cells are irradiated lethally
(C)	Radiotherapy	(III)	Agents which convert normal cells to cancerous cells
(D)	α-interferon	(IV)	Useful to detect cancers of the internal organs

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-II, B-III, C-IV, D-I
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-IV, C-II, D-I

[Contd...

96. Upon denaturation, a protein loses its:
(1) primary structure
(2) peptide bonds
(3) sequence of amino acids
(4) three dimensional shape

97. The term used to refer to the use of bio-resources by multinational companies without proper authorization is**:**
(1) Bio-exploitation
(2) Bio-patenting
(3) Biotechnology
(4) Biopiracy

98. Consider the following statements.
I. There is always a time lag between the infection and appearance of AIDS symptoms.
II. The time period for the AIDS symptoms may vary from a few months to many years, usually 5-10 years.

Choose the **correct** option.
(1) Statement I is true, but II is false
(2) Statement I is false, but II is true
(3) Both statements I and II are true
(4) Both statements I and II are false

99. The integration of natural science and organisms for products and services is the definition of biotechnology given by:
(1) Herbert Boyer
(2) Stanley Cohen
(3) National Center for Biotechnology Information
(4) European Federation of Biotechnology

100. What is the main reason why insulin cannot be administered orally to diabetic patients?
(1) It is too large to be absorbed by the intestine
(2) It requires an alkaline pH to become active
(3) It is toxic to the stomach lining
(4) It is a protein that would be digested in the gastrointestinal tract

■■■

SPACE FOR ROUGH WORK

Kindly Share Your Feedback For this Paper

अपनी भावनाओं को समझें, तनाव और उलझनों से बाहर निकलें।

आप अकेले नहीं हैं – आपकी मदद के लिए PW प्रेरणा काउंसलर हमेशा तैयार हैं।

प्रेरणा हेल्पलाइन: +91 99905 00122

Discover Your Path to Emotional Wellness – Speak with a PW Prerna Counsellor Today!

PRERNA Helpline: +91 99905 00122



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if